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Substance use patterns among youth seeking help for mental illness: A latent class analysis

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ABSTRACT

Background: Substance use is common among youth, with emerging adulthood a high-risk period for developing both substance use disorder (SUD) and mental illness. Youth receiving mental health treatment have higher rates of substance use than their peers, providing an opportunity for early intervention for SUDs. To facilitate this, a better understanding of substance use patterns among help-seeking youth is needed.
Objectives: We employed exploratory and confirmatory latent class analyses (LCA) to identify discrete classes of youth attending mental health services based on their substance use patterns, and assessed differences between groups in demographics, quality of life (QOL) and psychiatric symptoms.
Methods: Participants were treatment-seeking youth (15 − 25 years) recruited from *headspace*, Australia's national network of youth-focused primary mental health services, in 2 cohorts (Study 1, n = 676, female = 67.8 %; Study 2, n = 295, female = 66.3 %). Measurements included demographics, lifetime and recent substance use, mental health symptomatology and QOL.
Results: Exploratory LCA (Study 1) revealed a four-class model of substance use: 1) current alcohol or no substance use (ALC), 2) current tobacco, alcohol, and cannabis use (TAC), 3) past polysubstance use, and 4) current polysubstance use. The current polysubstance use group reported more psychological distress than the ALC group and lower QOL than youth without polysubstance use (ALC and TAC). Confirmatory LCA (Study 2) identified four similar classes, however no differences between groups in distress or QOL were observed.
Conclusion: Findings identify clinically-significant substance use patterns among youth accessing mental health care, with implications for development of early interventions to address substance use in this risk-enriched population.

1. Introduction

Adolescence and young adulthood are peak periods for initiation and use of alcohol and other drugs (AOD) (Degenhardt et al., 2016; SAMHSA, 2022). Estimates from 2019 showed that, in the past year, almost one third of Australians aged 18 − 24 had used illicit drugs and almost half of those aged 12 − 17 used alcohol (Institute, 2023). Similarly, in the USA, 40.9 % of 18 − 25 year olds reported past year illicit drug use, higher rates than for any other age group (SAMHSA, 2022). Early onset substance use is associated with adverse consequences

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including development of substance use disorders (SUDs), overdose, school dropout and physical and mental health problems (Degenhardt et al., 2016; Degenhardt et al., 2013; Erskine et al., 2015). Indeed, substance use is among the leading drivers of the overall burden of disease in 15 − 24 year olds (AIHW, 2023). Polydrug use is associated with particularly poor outcomes, including increased mortality, poor treatment response, and impaired functioning (Crummy et al., 2020).

Youth in mental health treatment have higher rates of substance use than their peers and are at greater risk of developing an SUD (Baker and Kay-Lambkin, 2016; Kessler et al., 2005; Kessler et al., 2011; Carney et al., 2017). Co-occurring mental illness and substance use increase risks for more severe psychiatric symptoms and poorer mental health treatment outcomes (Mammen et al., 2018; Gobbi et al., 2019). Early intervention may reduce substance use and related harms, yet youth are often hesitant to engage with services for substance-specific care (Scott et al., 2012; Reavley et al., 2010). However, many young people do present for mental health treatment, which, given common comorbidity with SUDs, presents an opportunity to access this high-risk population, often years before they would present for substance use treatment (Chapman et al., 2015; Blanco et al., 2015).

Identification of groups based on patterns of substance use is important for understanding clinically-relevant individual differences and prognostic profiles to guide intervention. A useful method to identify groups that may be at particularly high risk of substance-related harms is latent class analysis (LCA). LCA is a person-centred method for examining the underlying relationships among multivariate categorical data (McCutcheon, 1987). It allows for modelling of a taxonomic structure of substance use by grouping individuals into mutually exclusive classes based on similar features. As LCA is model-based, the generalisability of a model from a given dataset to a population can be assessed through cross-validation with a separate sample (Collins et al., 1994; Whitesell et al., 2006).

Several studies have employed LCA to categorise substance use patterns in youth. A systematic review of 70 cluster-based analyses (including LCA) of adolescent (11–18 years old) substance use in both general populations and those with co-occurring mental illness identified commonly occurring clusters across studies, including (a) low use, (b) single or combined use of alcohol, tobacco or cannabis, (c) moderate general polysubstance use including alcohol, tobacco and cannabis with or without other substances, and (d) high polysubstance use (Halladay et al., 2020). Polysubstance use in adolescents is consistently associated with poorer outcomes compared to low or no use, including negative psychosocial effects, higher psychological distress and worse general health (Kulis et al., 2016; Bohnert et al., 2014). A systematic review of 20 LCAs of substance use in non-treatment seeking young adults (18–25 years) similarly identified four commonly occurring clusters in 14 out of 20 studies, including (a) low use, (b) light alcohol and tobacco use, (c) alcohol and tobacco use, and (d) heavy use/polysubstance use (de Jonge et al., 2022). Predictors of belonging to the heavy use class included older age, male gender, and a history of mental illness.

Although LCA studies have been conducted in the USA and Europe to identify substance use patterns among help-seeking youth (Halladay et al., 2020), little is known about youth presenting to primary mental health care services, or Australian help-seeking youth. Given evidence for high rates of comorbidity between mental illness and substance use problems among help-seeking youth AIHW, 2023, Baker and Kay-Lambkin, 2016, Tervo-Clemmens et al., 2024; youth engaging with *headspace* – Australia's nationwide network of youth-oriented primary mental healthcare services – represent a unique population for understanding patterns of youth substance use and how they relate to demographics, psychiatric symptoms, and overall quality of life (QOL). *Headspace* was established in 2006 to provide youth aged 12 − 25 years with early intervention-oriented mental healthcare as well as co-located AOD services (Purcell et al., 2015; McGorry et al., 2007). The low barrier model of *headspace* is relatively unique internationally, presenting opportunities for proactive intervention for substance-related problems. Consistent with the documented reluctance of youth to present for specialist substance use treatment, most young people access *headspace* for mental health treatment (72.7 %), with only 3.1 % identifying substance use as their primary concern (Rickwood et al., 2015). Despite low rates of specialised substance use treatment, more than 50 % of youth presenting for mental health treatment at *headspace* endorse problematic substance use (Purcell et al., 2015b). Yet, insight into substance use patterns and associated characteristics in this help-seeking population remains limited, presenting a challenge for developing early interventions and treatments for SUDs in this risk-enriched population.

We aimed to investigate substance use characteristics of youth aged 15 − 25 years presenting for *headspace* services by identifying empirically-defined subgroups of individuals through exploratory LCA (E-LCA) based on both lifetime and recent (i.e., past three-month) substance use. This contrasts with most studies which employ either a single timepoint (lifetime or recent) or frequency of use as indicator variables. Our latent class models were then cross-validated through confirmatory LCA (C-LCA) using a separate sample of similar youth presenting to *headspace* services. We also assessed differences in demographic, psychological, substance use risk, and QOL features between substance use classes to better understand the characteristics of youth who may be at risk for developing SUDs and other substance-related harms.

2. Materials and methods

Data were derived from two sources. For the E-LCA, we conducted a secondary analysis of data from the Transitions Study (Study 1; see 25, 28). Briefly, Transitions established a cohort of youth seeking help for mental illness via *headspace* to test the clinical staging model of mental disorders. The C-LCA was a primary analysis of the Mood And Substance use in Help-seeking yoUng People (MASH-uP) Study (Study 2), described below.

2.1. Participants and procedures

2.1.1. Study 1: Transitions

Participants were 802 youth aged 12 − 25 years (inclusive) presenting at *headspace* centres in Sydney (n = 2) or Melbourne, Australia (n = 2) between January 2011 and August 2012. All youth who were English-speaking, able to provide informed consent and not acutely suicidal were invited to participate. After obtaining written informed consent, clinical measures were administered as structured interviews (details below). Participants then completed self-report measures with an iPad or laptop (details below). The study was approved by Human Research Ethics Committees (HRECs) at the University of Sydney and the University of Melbourne. To facilitate comparison with the MASH-uP sample, we restricted analyses to participants aged 15 − 25 years; those aged 12 − 14 years ($n = 106$) and/or with any missing data on the World Health Organisation Alcohol, Smoking, and Substance Involvement Screening Test (WHO-ASSIST; $n = 20$) from which the LCA indicator variables were derived were excluded, resulting in a final sample of 676 participants.

2.1.2. Study 2: MASH-uP

Participants were 295 youth aged 15–25 years seeking treatment at one of four *headspace* centres (2 overlapping with Study 1 sites) in Melbourne, Australia between July and October 2018. Participants were recruited from waiting rooms. All youth who were English-speaking, currently receiving care at *headspace* and able to provide informed consent were eligible. Participants provided written informed consent. All questionnaires were presented via iPad. The study was approved by the University of Melbourne HREC (ID#1851047).

2.2. Measures

World Health Organisation Alcohol, Smoking, and Substance

Involvement Screening Test (WHO-ASSIST). The WHO-ASSIST (Humeniuk et al., 2008) assessed lifetime and past three-month use and problems related to use of: alcohol, tobacco, cannabis, sedatives, amphetamine-type stimulants (including ecstasy), hallucinogens, cocaine, opioids and inhalants. It was administered by structured interview in Study 1 and self-report in Study 2. For each substance, we added the scores from questions 2 through to 7 to calculate an ASSIST risk score indexing substance-related risk (Humeniuk et al., 2010). Participants scoring between 4 and 26 (11 and 26 for alcohol) are at moderate risk of health and other problems; participants scoring $\geq$ 27 are at high risk.

Abbreviated WHO Quality of Life measure (WHOQoL-BREF-1). QOL was assessed using item 1 from the WHOQoL-BREF, as previously employed (Murphy et al., 2000). Participants rated their past four weeks' QOL on a five-point Likert scale. WHOQoL-BREF-1 was administered by structured interview in Study 1 and self-report in Study 2.

Demographic questionnaire. Characteristics assessed included age, sex at birth, country of birth, languages spoken at home, current living arrangements, highest level of education, current employment, marital/relationship status, and sexual orientation. Race/ethnicity was not assessed.

2.2.1. Emotional distress and internalising symptoms

Kessler 10 Psychological Distress Scale (K10). The *K*10 (Kessler et al., 2002) was used to measure psychological distress in Study 1. Respondents rated negative emotional states (past four weeks) on a five-point Likert scale. Items were summed to a total score, with higher scores indicating greater distress.

7-item Generalised Anxiety Disorder Scale (GAD-7). The GAD-7 (Spitzer et al., 2006) was used in Study 1 to measure generalised anxiety symptoms (past two-weeks) using a four-point Likert scale. Items were summed to generate a total, with higher scores indicating greater anxiety severity.

*Depression, Anxiety, and Stress Scales (DASS-21).*The DASS-21 (Lovibond and Lovibond, 1995) was used in Study 2 to examine past week depression, anxiety, and stress. Participants rated each item on a four-point Likert scale. Total scores for depression, anxiety, and stress were generated from summed totals for each of the 3 subscales, multiplied by 2 to equate to the 42-item version. Higher scores indicate greater depression, anxiety or stress.

2.3. Statistical analyses

Descriptive analyses and comparisons of the two samples were performed using IBM SPSS Statistics Version 27. The LCAs were performed using the poLCA software package (Linzer and Lewis, 2011; Linzer and Lewis, 2014) in RStudio Version 2022.02.2 (RStudio Team, 2016), an integrated development environment for R (R Core Team, 2018).

First, E-LCAs were conducted to identify potential models of youth substance use patterns in Study 1. The most parsimonious, theoretically-supported model was cross-validated using Study 2 (i.e., the C-LCA). The indicator variables included in both the E-LCAs and C-LCAs were categorical measures of substance use from the WHO-ASSIST. Lifetime and past three-month use data were recoded into trichotomous variables for each substance (i.e., 1 = *no past use*, 2 = *lifetime use but not in the past three months*, 3 = *use in the past three months*). Consistent with the approach taken in the ASSIST-3, we operationalised recent use as being use within the past 3 months, to identify substance use most likely to be relevant to participants' health and wellbeing at the time of assessment (WHO ASSIST Working Group. The Alcohol, 2002).

To identify the most parsimonious model in the E-LCA, an independence model was fit to the data first (i.e., one latent class), to which progressive classes were added, until the fit worsened. Goodness-of-fit was evaluated using the Bayesian Information Criterion (BIC) (Schwarz, 1978) and Akaike's Information Criterion (AIC) (Akaike, 1973) indices. The Pearson's χ^2 goodness-of-fit and G^2 likelihood ratio statistics were considered but to a lesser extent, as both are sensitive to violations of distributional assumptions (Linzer and Lewis, 2011). For all goodness-of-fit criteria, lower values indicated better balance between parsimony and model fit (Lanza and Rhoades, 2013).

In poLCA, maximum likelihood estimates of model parameters are found using expectation–maximization and Newton-Raphson algorithms, which often only determine local, rather than the desired global, maxima (McLachlan and Krishnan, 1997). Thus, the number of repetitions with different parameter estimates was increased to 10 to ensure that the global maximum log-likelihood solution was identified (Linzer and Lewis, 2011). The maximum number of algorithm iterations was also increased to 50,000 to ensure convergence on a model before the analysis terminated. All other elements were kept as the default (Linzer and Lewis, 2014).

Once the most parsimonious model was determined, classes were interpreted based on the high conditional probabilities for each indicator variable. Finally, the C-LCA was performed with Study 2 data by constraining the number of classes based on the E-LCA (Schmiege et al., 2018).

To examine demographic, psychological, substance risk, and QOL correlates of latent classes for the E-LCA and C-LCA, we conducted bivariate analyses (e.g., χ^2, analysis of variance [ANOVA], Kruskal-Wallis test) and post-hoc tests with Bonferroni correction with an alpha of 0.05.

3. Results

3.1. Sample characteristics

Table 1 presents demographic characteristics of the samples. Age was similar across samples, however, more young adults (20–25 years) participated in Study 2 (52.2 %, n = 295) than Study 1 (41.6 %, n = 281). Approximately two-thirds of participants were female across studies. Almost all participants were Australian-born and spoke English at home. Almost two-thirds were enrolled in education. Less than half (44.9 %) of Study 1 participants and one third (32.9 %) of Study 2 participants were employed, predominantly in part-time work.

3.2. Mental health characteristics

In Study 1, the mean *K*10 score (29.5, $SD = 9.5$, $n = 657$) indicated that on average, participants were likely to have a moderate to severe mental disorder. More than two-thirds (69.7 %) met the cut-off for being likely to have at least a 'moderate mental disorder'. The mean GAD-7 score (10.1, $SD = 6.0$, $n = 657$) was consistent with 'moderate anxiety'.

In Study 2, mean DASS-21 scores were consistent with severe depression ($M = 21.5$, $SD = 11.0$, $n = 288$) and anxiety ($M = 17.0$, $SD = 9.6$, $n = 290$), and moderate stress ($M = 20.7$, $SD = 9.1$, $n = 286$).

The mean WHOQoL-1 score was 3.1 ($SD = 1.1$, $n = 652$) and 3.3 ($SD = 1.0$, $n = 276$) in Studies 1 and 2 respectively, indicating a QOL that is 'neither poor nor good'. QOL was rated poor or very poor by 28.5 % and 22.0 % of participants in Studies 1 and 2, respectively.

3.3. Substance use patterns

Lifetime and past three-month substance use is reported in Table 2.

Across samples, the most used substances were alcohol, tobacco, and cannabis. Compared to Study 2, participants from Study 1 reported higher lifetime use of alcohol, tobacco, cannabis, and amphetamine-type stimulant use and lower sedative and opioid use. Approximately three-quarters of participants across studies reported alcohol use in the past three months, half reported tobacco use, and one-third cannabis use. Past three-month use of sedatives, cocaine, opioids and inhalants was higher in Study 2 than Study 1. Only 3.0 % (n = 20) of Study 1 and 1.2 % (n = 3) of Study 2 participants reported ever injecting drugs.

Table 1
Summary of demographic characteristics.

Variable	Study 1 (n = 676)			Study 2 (n = 295)		
	%	n	Missing	%	n	Missing
Sex assigned at birth (Female)	67.8	458	0	66.3	193	4
Age, M (SD)	19.0 (2.8)		0	19.5 (2.9)		0
Country of birth (Australia)	90.8	601	14	88.8	262	0
Language at home (English)	98.2	649	15	93.1	270	5
Living at home	67.2	444	15	70.4	207	1
Engaged in education (total)	64.0	423	15	61.9	178	6
School	27.9	184	16	32.5	94	6
University/College/TAFE/ Other	35.9	237	16	29.4	85	6
Engaged in employment (total)	44.9	297	15	32.9	95	6
Full-time	9.7	64	15	6.9	20	6
Part-time	35.2	233	15	26.0	75	6
Not in education, employment or training	18.8	124	15	21.1	61	6
Relationship status			15			1
Single/never married	60.1	397		65.0	191	
Partnered (<3 months)	7.9	52		6.1	18	
Partnered (3 months – 2 years)	28.0	185		18.3	54	
Married/defacto	3.5	23		8.2	24	
Other	0.6	4		2.4	7	
Sexual orientation			17			4
Heterosexual	74.1	488		57.7	168	
Bisexual	14.3	94		25.1	73	
Same-sex attracted	5.5	36		6.5	19	
Don't know/don't want to say	6.2	41		10.7	31	
Highest level of education			15			10
Less than year 10	23.3	154		13.0	37	
Year 10	25.9	171		24.6	70	
High school/Year 12	31.9	211		36.1	103	
Trade/apprenticeship	1.4	9		N/A	N/A	
Certificate/diploma/ advanced diploma	11.3	75		17.2	49	
University degree or higher	6.2	41		9.1	26	

TAFE: Technical and Further Education.

Table 2
Lifetime and past three-month substance use (ASSIST) of participants from Studies 1 and 2.

	Study 1 (n = 676)	Study 2 (n = 295)	Test	
	% (n)	% (n)	χ^2	p
Lifetime				
Alcohol	91.7 (620)	84.1 (248)	**12.7**	**0.001**
Tobacco	72.0 (487)	55.3 (163)	**26.2,**	**< 0.001**
Cannabis	56.4 (381)	49.5 (146)	**3.9**	**0.048**
Sedatives	16.1 (109)	29.8 (88)	**23.9**	**< 0.001**
Amphetamine-type stimulants (including ecstasy)	30.0 (203)	22.4 (66)	**6.0**	**0.014**
Hallucinogens	20.4 (138)	17.6 (52)	1.0	0.314
Cocaine	17.6 (119)	14.9 (44)	1.1	0.303
Opioids	6.7 (45)	14.2 (42)	**14.5**	**< 0.001**
Inhalants	10.2 (69)	10.8(32)	0.1	0.764
Other	4.3 (29)	4.7 (14)	0.1	0.751
Past 3-month				
Alcohol	75.7 (512)	76.3 (225)	0.0	0.859
Tobacco	53.0 (358)	44.1 (130)	**6.5**	**0.011**
Cannabis	32.5 (220)	32.9 (97)	0.0	0.918
Sedatives	6.2 (42)	18.3 (54)	**33.7**	**< 0.001**
Amphetamine-type stimulants (including ecstasy)	10.7 (72)	11.2 (33)	0.6	0.805
Hallucinogens	6.2 (42)	8.1 (24)	1.2	0.274
Cocaine	3.7 (25)	9.8 (29)	**14.7**	**< 0.001**
Opioids	2.8 (19)	6.8 (20)	**8.4**	**0.004**
Inhalants	2.4 (16)	5.4 (16)	**6.0**	**0.014**
Other	0.9 (6)	0.3 (1)	0.9	0.353

Note. Values in **bold** denote a significant difference at $p < 0.05$ between Studies 1 and 2.

3.4. Exploratory latent class analysis (E-LCA)

In the E-LCA, the best-fitting model was identified by comparing fit criteria as the number of classes increased. All values decreased as classes increased, indicating improved model fit, until a four-class model was estimated, which resulted in a slight increase in BIC but a reduction in all other criteria (Table 3). A five-class model was rejected due to a further increase in BIC. As BIC is a conservative criterion known to occasionally underestimate the number of classes (Dziak et al., 2014), we examined conditional item response probabilities for the three- and four-class models. Inspection of the four-class model revealed two classes characterised by high probabilities of past three-month or lifetime polysubstance use, whereas in the three-class model these two clinically-relevant classes were collapsed into one. The four-class model was thus the most theoretically and clinically consistent and was chosen as the final model.

Classes were labelled based on proportion of participants endorsing lifetime and past three-month use of each substance. Patterns were suggestive of: (1) Current alcohol or no substance use (ALC; 44.5 % of the total sample), characterised by moderate probabilities of alcohol use (52.9 %) in the past three months and low probabilities of lifetime use of other substances; (2) Current tobacco, alcohol, and cannabis use (TAC; 33.9 % of the total sample), with high probabilities of lifetime tobacco (96.6 %), alcohol (100 %) and cannabis use (89.2 %), moderate to high use of these substances in the past three months (77.3 %, 92.4 % and 46.9 % respectively), and low use of other substances; (3) Current alcohol, tobacco, and cannabis use, and past polysubstance use, (hereafter referred to as past polysubstance use; 10.8 % of the total sample), also characterised by moderate to high probabilities of past three-month tobacco (76.8 %), alcohol (83.6 %) and cannabis use (53.8 %) and moderate-to-high probabilities of lifetime, but not recent, use of other drugs (including amphetamine-type stimulants (100 %), cocaine (65.2 %) and hallucinogens (63.6 %)); and (4) Current polysubstance use (10.8 % of the total sample), characterised by higher probabilities of having used alcohol (98.5 %), tobacco (96.5 %), cannabis (87.3 %) and a range of other substances (including amphetamine-type stimulants (77.6 %) and hallucinogens (47.6 %)) in the past three months.

3.5. Confirmatory latent class analysis (C-LCA)

A C-LCA using the Study 2 sample was used to validate the E-LCA. Using a dual sample approach (Schmiege et al., 2018), we constrained our model within a narrower range based on the four-class E-CLA (i.e., 3–5 classes). All fit criteria decreased from the three- to four-class model, with the exception of BIC (Table 4). After review of the conditional probabilities, the four-class model was chosen as the most satisfactory solution. Assessment of local independence for the E-LCA and C-LCA is included in supplemental information (see Tables S1 and S2), as are classification diagnostics (Table S3).

Substance use characteristics of the C-LCA four classes were generally consistent with those from the E-LCA. Patterns of lifetime and past three-month use across classes in the two models are depicted in Fig. 1 (see Table S4 for conditional probabilities for substance use variables). As with the E-CLA, the first class (current alcohol or no substance use [ALC]; 39.7 % of the total sample) was characterised by moderate probabilities of past three-month alcohol use (52.7 %) with low use of other substances. Class 2 (current tobacco, alcohol, and cannabis use [TAC]; 38.3 % of the total sample) demonstrated moderate to high probabilities of past three-month tobacco (66.0 %), alcohol (90.9 %) and cannabis use (41.7 %) and low use of other drugs. Class 3 (past polysubstance use; 11.5 %) was characterised by moderate to high probabilities of alcohol (88.8 %) tobacco (63.4 %) and cannabis use (54.4 %)

Table 3
Goodness-of-fit statistics for *exploratory* models of substance use according to class enumeration (n = 676).

No. of classes	LL	AIC	BIC	G^2	χ^2	df	Entropy‡
1	−4194.27	8428.539	8518.863	2795.219	30,693,344	656	6.20
2	−3494.528	7071.056	7256.22	1395.736	96936.38	635	5.22
3	−3321.77	6767.541	7047.545	1050.22	32449.00	614	4.95
4	**−3272.73**	**6711.459**	**7086.303**	**952.1392**	**29343.89**	**593**	**4.87**
5	−3235.631	6679.262	7148.946	877.9422	24241.81	572	4.80

Note. **Bold** text indicates model selected. ‡Maximum entropy (representing complete dispersion, or equal probability for outcomes across every cell) = 10.99. Minimum entropy (representing complete concentration of probability on one cell) = 0.00. *LL*, Log-Likelihood; *AIC*, Akaike's Information Criterion; *BIC*, Bayesian Information Criterion.

Table 4
Goodness-of-fit statistics for *confirmatory* model of substance use according to class enumeration (n = 276).

No. of classes	LL	AIC	BIC	G^2	χ^2	df	Entropy‡
3	−1520.196	3164.392	3392.984	737.4302	12,480,749	233	5.16
4	**−1489.16**	**3144.321**	**3450.34**	**675.3593**	**168,219**	**212**	**5.06**
5	−1470.158	3148.316	3531.761	637.3545	161139.7	191	4.99

Note. **Bold** text indicates model selected. ‡Maximum entropy (representing complete dispersion, or equal probability for outcomes across every cell) = 10.99. Minimum entropy (representing complete concentration of probability on one cell) = 0.00. *LL*, Log-Likelihood; *AIC*, Akaike's Information Criterion; *BIC*, Bayesian Information Criterion

in the past three months, with moderate probabilities of lifetime, but not recent, use of other drugs. Finally, Class 4 (current polysubstance use, 10.5 % of the total sample) showed high probabilities of current tobacco, alcohol and cannabis use (≥97 %) with moderate use of other drugs.

3.6. Class differences in demographic and mental health characteristics

Characteristics by class from Study 1 and 2 are presented in Table 5. In the E-LCA, groups differed by age and sex at birth; the two polysubstance use classes were older, on average, and had proportionally more males than did ALC and TAC classes. Current polysubstance use also reported more distress and anxiety than youth with or no substance use, and a lower QOL than youth without polysubstance use (i.e. both ALC and TAC groups). Similar patterns were observed in the C-LCA, but only the effect of age reached statistical significance.

3.7. Class differences in substance use characteristics

Mean ASSIST risk scores for each class are shown in Table S5. The majority of participants in the TAC, past polysubstance use, and current polysubstance use groups had at least one risk score in the moderate-to-high range (Table 6). The current polysubstance use group had more substances, on average, for which scores were in the moderate-to-high range than other groups (3.5 and 4.2 substances in Studies 1 and 2 respectively). Compared to the alcohol only group, the TAC and past polysubstance use groups also had more substances with scores in the moderate-to-high range. The frequency of alcohol, tobacco and cannabis use across classes is presented in Table S6.

4. Discussion

We used E-LCA and C-LCA to investigate patterns of substance use across two samples of youth seeking treatment for mental illness. Four distinct subgroups were identified in both analyses: two larger classes characterised by (a) predominantly alcohol or no substance use and (b) tobacco, alcohol and cannabis use; and two smaller classes characterised by polysubstance use, differentiated by high probabilities of (c) prior (lifetime but not current) or (d) current polysubstance use. Importantly, these classes were consistent across both samples, despite these being collected some 6 years apart. We also identified clinically meaningful differences between groups in substance use risk, and on key demographic and psychological indicators in the larger Study 1 sample, with the current polysubstance use class being older, with higher levels of psychological distress and lower QOL than youth with ALC and/or TAC use. In both studies, most young people in the TAC group and almost all those in the two polydrug groups had moderate-to-high risk use of at least one substance, indicating widespread clinically significant substance use in young people seeking primary mental health care.

Findings are broadly consistent with those of a systematic review of 70 cluster-based studies of adolescent who use substances, which identified several commonly occurring clusters, including (a) no or low substance use, (b) single or dual use of alcohol, tobacco or cannabis, (c) moderate general multi-drug use, and (d) high multi-drug use (Halladay et al., 2020). Similarly, a recent systematic review of 20 LCAs of substance use in non-treatment seeking young adults identified four commonly occurring clusters: (a) low level engagers, (b) light alcohol and tobacco use, (c) alcohol and tobacco use, and (d) heavy use/polysubstance use (de Jonge et al., 2022). Notably, we did not identify a "no/low substance use" class (although the ALC class had only a moderate probability of alcohol use in the past 3 months; approximately 50 % reported no use). Of note, most studies in Halladay et al. (Halladay et al., 2020) and all of those in de Jonge et al. (de Jonge et al., 2022) were in the general population rather than treatment-seeking youth – a risk-enriched population. In addition, few studies in Halladay and colleagues (Halladay et al., 2020) were conducted outside the US, or examined substance use in adolescents with mental illness. To our knowledge, no prior study has focused on substance use in youth seeking treatment for mental illness. The present study provides a model of substance use patterns in treatment-seeking Australian youth that is cross-validated across two samples, strengthening generalisability.

This study also provided a unique perspective on youth substance use patterns by combining lifetime and recent use within a single model. Previous studies have primarily investigated lifetime and recent substance use separately and have generally identified a single polysubstance use class characterised by use of alcohol, tobacco, cannabis and other drugs (Whitesell et al., 2006; Halladay et al., 2020; de Jonge et al., 2022; Su et al., 2018; Riehman et al., 2009; White et al., 2013). By contrast, we identified two distinct classes of polysubstance use differentiated by either past or current use of multiple substances. These classes were older than subgroups with more limited use in both Studies 1 and 2, consistent with prior research and the increased prevalence of polysubstance use in young adults compared to adolescents (Steinhoff et al., 2021). In the E-LCA, both polysubstance use classes also had proportionally more males, which is consistent with studies reporting

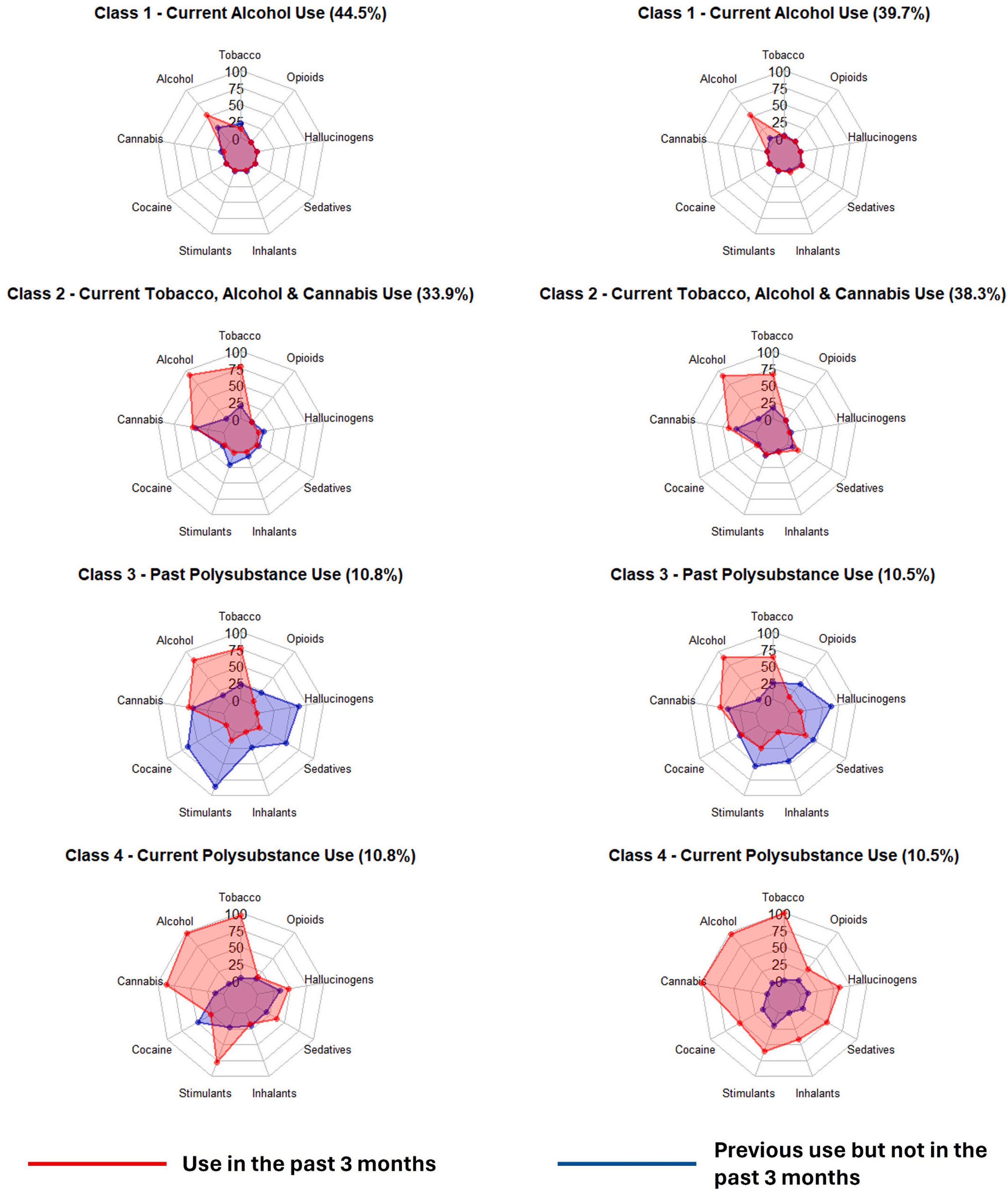

Fig. 1. Radar plots depicting the conditional probabilities of participants endorsing lifetime and past three-month use of each substance for each class in the exploratory and confirmatory latent class analyses using Study 1 (E-LCA; left panel) and Study 2 samples (C-LCA; right panel).

that males are more likely to use multiple substances (John et al., 2018; Chan et al., 2020). Conversely, only the current polysubstance use class had higher psychological distress and poorer QOL compared to ALC users in the E-CLA. This suggests that mental health and QOL may improve after discontinuation of polysubstance use.

While our analyses were based on substance use rather than SUD, rates of moderate-to-high ASSIST scores – where moderate scores are indicative of harmful or hazardous use and high scores likely represent

Table 5
Demographic, psychological, and quality of life variables associated with the four classes of the exploratory and confirmatory models of substance use.

Variable	ALC	TAC	PAST POLY	CURRENT POLY	Test
% (n) of participants					
E-LCA (total = 676)	44.5 (301)	33.9 (229)	10.8 (73)	10.8 (73)	–
C-LCA (total = 295)	39.7 (117)	38.3 (113)	11.5 (34)	10.5 (31)	–
Age, *M* (*SD*)					
E-LCA	18.2 (2.7)[a]	19.0 (2.6)[b]	21.4 (2.4)[c]	20.1 (2.8)[d]	**F = 32.6, p < 0.001**
C-LCA	18.4 (2.7)[a]	19.7 (2.8)[b]	22.2 (2.3)[c]	20.3 (2.6)[d]	**F = 18.7, p < 0.001**
Sex, % female					
E-LCA	72.4	68.1	60.3	54.8	**χ^2 = 10.5, p = 0.015**
C-LCA	71.3	65.2	58.8	60.0	χ^2 = 2.7, p = 0.434
Psychological distress, *M* (*SD*)					
E-LCA					
*K*10 score	28.6 (9.3)[a]	29.6 (9.7)[a,b]	30.0 (9.3)[a,b]	32.4 (9.2)[b]	**F = 3.3, p = 0.020**
GAD-7 score	9.7 (5.8)[a]	9.8 (6.0)[a,b]	11.1 (6.4)[a,b]	12.1 (5.6)[b]	**F = 3.8, p = 0.010**
C-LCA					
DASS-21 (Depression)[ǂ]	21.3 (10.94)	21.6 (10.43)	18.6 (10.94)	24.9 (12.67)	F = 1.8, p = 0.151
DASS-21 (Anxiety)[ǂ]	16.5 (9.4)	17.5 (9.6)	15.7 (9.3)	18.7 (11.0)	F = 0.8, p = 0.526
DASS-21 (Stress)[ǂ]	19.9 (8.8)	20.8 (8.7)	21.2 (9.6)	22.73 (11.1)	F = 0.8, p = 0.480
Quality of life (WHOQOL-1), *M* (*SD*)					
E-LCA	3.2 (1.0)[a]	3.0 (1.1)[a]	3.0 (1.2)[a,b]	2.6 (1.1)[b]	**F = 6.4, p < 0.001**
C-LCA	3.3 (1.0)	3.2 (1.0)	3.4 (1.1)	3.1 (0.9)	F = 0.8, p = 0.511

Note. Bold text indicates significant differences.
For the bivariate analyses, significant differences identified through post-hoc tests with Bonferroni correction are indicated through superscripts; groups that are not significantly different are assigned a common letter.
ALC: current alcohol or no substance use; TAC: current tobacco, alcohol, and cannabis use, PAST POLY: past polysubstance use, CURRENT POLY: current polysubstance use.
ǂ DASS-21 scores multiplied by 2 to facilitate comparison to the full 42-item version of the scale.

substance dependence (WHO ASSIST Working Group. The Alcohol, 2002) – were high. Indeed, more than 75 % of participants in 3 of the 4 groups (TAC, past polysubstance use, current polysubstance use; together comprising 55–60 % of the overall samples) had moderate-to-high ASSIST scores for at least one substance. This was higher in the polysubstance use groups. This suggests that a large proportion of young people seeking primary health care have existing SUDs, and that risk of SUD tracks with the amount of polysubstance use. Current polysubstance use, in particular, was a marker for extremely high rates of moderate-to-high risk use of at least one substance (>97 %). Moreover, the current polysubstance use group were also likely to have moderate-to-high risk scores for several substances (3 to 4 substances on average).

Table 6
Measures of substance use risk in Studies 1 and 2.

	ALC	TAC	PAST POLY	CURRENT POLY	χ^2	p
% (n) of participants at moderate-to-high risk of substance use problems						
Study 1	16.9 (51)[a]	79.5 (182)[b]	86.3 (63)[b,c]	97.3 (71)[c]	**312.2**	**0<.001**
Study 2	14.5 (17)[a]	76.1 (86)[b]	88.2 (30)[b,c]	100 (31)[c]	**138.6**	**0<.001**
Average no. of moderate-to-high risk score substances/person, M (SD)						
Study 1	0.2 (0.4) [a]	1.4 (0.9) [b]	1.7 (1.3) [b]	3.5 (2.0) [c]	**244.2**	**0<.001**
Study 2	0.2 (0.4) [a]	1.6 (1.2) [b]	2.4 (1.6) [b]	4.2 (1.7) [c]	**120.7**	**0<.001**

Note. Bold text indicates significant differences in the Chi Square test and Kruskall-Wallis test.
Significant differences identified through post-hoc tests with Bonferroni correction are indicated through superscripts; groups that are not significantly different are assigned a common letter.
ALC: current alcohol or no substance use; TAC: current tobacco, alcohol, and cannabis use, PAST POLY: past polysubstance use, CURRENT POLY: current polysubstance use.

Findings have clear implications, given that most youth presenting to headspace decline specialised substance use treatment (Purcell et al., 2015), suggesting a need to consider integrated care models (whereby substance use treatment is integrated within broader mental health care). The high levels of clinically significant substance use observed in our samples also suggest a need for targeted training programs for mental health clinicians to ensure that substance use is considered as part of standard treatment. As shown in Table 6, findings suggest that current cannabis use as well as current and past polysubstance use indicate an extremely high likelihood of substance use disorder in youth presenting for mental healthcare. This knowledge could inform safety and treatment planning even in the absence of a comprehensive assessment of substance use problems at intake, which youth presenting for mental healthcare (and declining substance specific care) may be unwilling to complete. Further, current polydrug use may be a marker for worse mental health symptoms and quality of life relative to young people with alcohol or no drug use, or use of cannabis, alcohol and tobacco, also highlighting the potential prognostic and treatment planning value of this information.

Further research on specific models of integrated care is urgently needed to address the unmet need identified in this study. This could potentially include brief early interventions for young people at lower risk levels with more intensive integrated treatments – including multidisciplinary group care where indicated – for those with existing SUDs. While integrated care has long been identified as best practice for people with comorbid psychiatric and substance use disorders, internationally implementation has proven challenging (McGinty and Daumit, 2020). There is little evidence on optimal integration approaches to guide treatment for youth with co-existing substance use and mental health difficulties.

This study is not without limitations. As a cross-sectional study, causal inferences cannot be made regarding associations between patterns of substance use, changes over time, and relationships between substance use and demographic, psychological, and QOL characteristics. There were some differences between Studies 1 and 2 that may have affected comparability between the samples: (a) Study 1 had a larger sample size than Study 2; (b) different measures of psychological distress were used in each study (GAD-7, *K*10 versus DASS-21); and (c) the ASSIST was administered in a self-report format in Study 2 and interviewer-administered format in Study 1. Despite the differences in the measures, we are confident that they were comparable on the basis of previous studies showing concordance between the DASS-21 anxiety/stress score and the GAD-7 Peters et al., 2021, Lee and Kim, 2019; the

*K*10 and the DASS-21 subscales (Peixoto et al., 2021), and the self-report and interviewer-administered versions of the ASSIST (McNeely et al., 2016). However, previous studies have also shown that self-administered instruments tend to generate higher rates of reporting of stigmatised behaviours (McNeely et al., 2016), potentially affecting our results. Although both studies comprised data from four comparably representative *headspace* sites in Australian metropolitan centres, the samples may have limited generalisability to rural populations. We also noted an over-representation of females in our samples, which is consistent with the higher presentation of young women for psychiatric care (GBD, 2019). Given there are higher levels of substance use among males (Degenhardt et al., 2013), this could have limited the extent of substance use detected in the current study.

In conclusion, the present study identified a closely similar four-class model of lifetime and past three-month substance use among youth seeking mental health care in two independent samples collected 6 years apart. In both samples, current and past polysubstance users were older than classes with lower probabilities of polysubstance use and had markedly higher levels of substance use risk. In Study 1, youth with current polysubstance use were more likely to be male, report higher levels of distress, and poorer QOL compared to other classes. There were no significant differences in demographic, psychological distress or QOL characteristics in Study 2, although similar patterns of results were observed. Highly similar rates of substance use were revealed in each of the classes across samples, suggesting relatively stable substance use characteristics in this population. Findings suggest there is a need for integrated care addressing high levels of clinically significant substance use in youth presenting for primary mental health treatment, particularly young people with polysubstance use who are at elevated risk of substance-related difficulties.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.addbeh.2025.108355.

Data availability

Data will be made available on request.

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